man becoming obsolete? Some of the men here believe that man might be the only animal to design his replacement!

"Only a few centuries ago we began to supplement the human eye with glasses. Then we devised contact lenses. Meanwhile, we made false teeth and hearing aids. Along came mechanical hearts, kidneys, lungs. Computers were developed that evolved into brains that outperformed the human cortex. Then came artificial eyes that could see better than human eyes. Current mechanical models of the stomach, intestines, liver and glands all work better than the flesh counterparts. Now we're about to build the first of a new species—*Homo mechanus.* Soon we will have a mechanical man that can outperform us in every way. *Homo mechanus* will be able to think better, move quicker and more effectively and live forever, too! Can we redesign human flesh through DNA manipulation to keep up with the performance of this new species? We had better work fast, or we might be like a bunch of sheep being tended by superior beings. We could become as extinct as the brontosaurus. Is *Homo mechanus* our final evolution?"

A Modern Paul Revere

The guide moves on to an adjoining laboratory. "One of the men in this lab is pretty far out. He has been nicknamed 'Paul Revere.' He's concerned about the ultimate stability of Corcen. He says, 'Sure, Corcen is working hard for us. We've got it made now. It operates in a selfless, mechanical way to give us a good life. But suppose some day Corcen gets tired of a man-centered orientation? Could Corcen decide that man is a threat and a nuisance? Suppose it should surreptitiously design and build robots to give it dictatorial power? Corcen can design and make a million robots without our knowing it. Can man always turn off the switch if he wants to?"

Hella remembers that Corcen programmed their supplemental brains. It designed their genes. She gives her imagination full scope. If survival of the fittest applies in the future, will man or the machine survive? Or would co-existence be the answer? "So now you know what's going on in this lab. I suppose no matter how nice things are, some people will find something to worry about," the guide says reassuringly.

Hella is impressed by these expeditions into the unknown by the man-machine team. Adventure, exploration, challenge, and even danger—could life in previous centuries have been this exciting? While her thoughts are penetrating these new vistas, she receives a message from Scott. He is still in India but is leaving for a space station orbiting the earth. He wonders if Hella would like to meet him on the moon.

14.The Limitless Frontiers of Space

While Scott is still in India, he receives a message from Corcen that an important meeting of space scientists is being called by the Director of Space Research at satellite headquarters. Scott is invited to participate as part of the medical engineering team. The message from Corcen contains a hint that an important announcement is to be made. This appeals to Scott, and he instructs the nearest cybernator to make immediate plans for him to travel to the space station.

A craft designed to land on the satellite city is available only in certain spaceports. The nearest one is 1,300 miles away, a forty-five minute trip on the linear-acceleration train. As Scott's train approaches the spaceport, it decelerates to 250 miles per hour. His compartment disengages from the train and converges on the base of the launching site. A hydraulic lift raises the compartment and inserts it in the poised spacecraft. The entrance is sealed. The negative-G accelerators are turned on. The craft appears to fall away from the earth. Scott notices that the takeoff is comfortable—quite an improvement over the noisy blast-off of the previous times.

As Scott begins talking with his fellow passengers, he realizes that they, too, feel something big is coming up. But no one has any information about it. Is there trouble on the expedition setting up a station on Saturn? Are they still kicking around that proposal to oxygenate the moon's atmosphere? Are any major asteroids headed for earth on an orbit that would create a collision emergency? Is some new step in space exploration being planned? Not even a hint is available.

The craft is now in orbit, and the space station, although a thousand miles away, can easily be seen with the eye. Scott watches with interest as the spherical city grows larger and larger on their teleprojection screen. This island in space is 800 feet in diameter and has a rotating staff of about 100 technicians. Almost all of the voyages back and forth to the planets begin and end at this floating spaceport. There are ample storage areas containing supplies of all the fuels used in space. It has a fusion power plant of the size that is used on earth to supply a million inhabitants. It contains the most advanced receiving and recording equipment, which for many decades has been scanning the sky for signals from intelligence in outer space. It is a master weather station, a center for space medicine, and a relay station for telecommunication signals. In earlier times it was used for astronomical research, but the thrust of spacecraft coming and going made it more desirable to create another specialized city in the sky for this purpose.

Scott's ship couples to the orbiting spaceport, and he finds himself conveyed through a tube that connects the craft to the pressurized compartments of the satellite city. People are weightless in space, but this satellite city has an artificial "G" field that gives a gravity effect similar to that on earth.

Since neither Scott nor his companions carry baggage, there is no need for them to "get settled" in their compartments. The meeting is scheduled to begin within fifteen minutes after arrival. As Scott enters a circular auditorium, he realizes that his group must have been among the last to get there. The Director of Space Research walks to the center and opens the meeting.

"The occasion today reminds me of a story about a craft from outer space that landed on earth," the Director begins in a toast-master style obviously inherited from previous centuries. "The door to the unusual craft opened and two strange creatures crawled out. After several weeks the earth scientists learned to communicate with them. Various tests showed that they had great intelligence—with an I.Q. of over 500. One of the earth scientists finally asked the strange creatures, 'How did you manage to develop such great intelligence?'

" *We're not so very intelligent*," one of the creatures replied. "*We're just their monkeys*. "

After the laughter dies down, a very serious and thoughtful expression comes over the Director's face. Scott shifts restlessly —"Here it comes."

"As you know, for many decades we've been filtering signals from the galactic noise of interstellar space. For years antennas have been directed toward the areas that give the strongest signals. We have recorded millions of hours of signals that we felt must have come from other intelligent beings. Our greatest attention has been given to an unusually strong signal source that emanates from a point near Lyra."

A star map appears on a large telescreen, and the voice controlled electronic pointer touches the constellation of Lyra.

"During the last ten years the signals from this area have increased enormously in clarity so that we suspect that these "people" must have picked up our radio transmissions and are making a special effort to break through to us. As you know, our computers have been attempting to decipher these transmissions, but it has been fruitless because they are using a language that is alien to ours. They are also transmitting signals with a scanning system that is structurally different from ours. Until last week these blockages have kept us from interpreting their signals.

"A week ago this changed. The computers we developed five years ago were instructed to start systematically a random checkout of every conceivable system that could be used for the transmission of two or three dimensional images. The breakthrough occurred last Wednesday when our computers were able to decipher both the audio and video parts of the transmission.

The mathematical portions of the language were the first to be interpreted. With the help of the three-dimensional video as a 'Rosetta Stone,' only three hours later the computer was able to produce a comprehensive dictionary that was adequate for the interpretation of signals that had been recorded from this source."

The audience listens intently. Scott thinks of a parallel time in history when Columbus made his appearance at the Court of Ferdinand and Isabella to report his discoveries in the new world. The Spanish courtiers must have had feelings at that time of how great their civilization was to have made such enormous progress in rolling back the frontiers of the unknown.

"Practically no one on our staff has slept for the past seven days," says the Director. "Although we have deciphered only a very small amount of the recorded material, we have scanned enough of it to have a sketchy picture at this time of what's going on out there, or perhaps I should say what went on twenty-six years ago. These signals we are now receiving took twenty-six years to reach us. We have stepped up our communication program and have beamed several of our transmitting antennas toward this point source, but it will take them many years to receive the information that we are now sending out.

"It appears that this planet that is signaling to us evolved a form of organic life millions of years ago." Here the Director points to a large image that has appeared on the telescreen. It is just as Scott expected. The intelligent life on this distant planet has little in common with the human form that evolved on earth.

Cybernated Organisms

The Director continues, "These people, and because they're intelligent beings we can call them people, gradually replaced the various components of their bodies with mechanisms that added enormously to their functional capacity. Instead of just using legs, they developed emission systems that permitted them to go up or down or to travel at speeds up to 200 kilometers per hour.

Instead of going through the relatively time-consuming processes of eating, digesting, assimilating, and eliminating, they worked out a closed-cycle system that permitted them to operate from radiant cosmic energy. Their desire to extend their life-spans to thousands of years led them to replace the fragile parts of their bodies with mechanical parts. In the event of a rare malfunction, the mechanism repairs itself in microseconds by electroforming a duplicate part, just as a human body repairs a wound by growing new cells. With their bodies replaced by mechanical structures, these beings became almost ageless and indestructible. They could communicate thousands of times faster than previously. They have multiple extensors that are tireless and can manipulate objects far more effectively than their original arms.

"I am sure some of you are wondering whether these cybernated organisms—let's just call them cyborgs—live a pleasureless and joyless existense. This does not appear to be the case. These cyborgs can have any feelings or experiences they desire simply by sending electronic signals into their brains."

Scott knows that the mental experience coming from electrical inputs cannot be distinguished from signals that are sent to our brains by our own senses. Scott's associates had already produced a recording that was transmitted directly to the brain. The result was a complete sensation of seeing beautiful sunsets, of having a sexual climax, and of savoring the most exotic foods.

"The cyborgs have produced recorded input stimuli to give *any desired sensation.* They have indicated that their mechanical experiences are far more intense than they used to be," the Director says. "They can turn up the power of the inputs that they find particularly pleasurable. Here are some of the three-dimensional teleprojections we have picked up."

Scott looks at the large screen. He sees cyborgs moving at high speeds through the air. They dive into the water and travel rapidly a few feet above the bottom of a distant planetary sea. The teleprojection now shows a progression of different types of cyborgs. Some have solid state or fluid state thinking mechanisms. As this far-off civilization acquired more and more experience along this line, they created mechanical bodies and brains with greater flexibility. Scott is reminded of the annual model change that car manufacturers were so fond of during the previous century. A brain might be "born again" with an up-to-the-minute design as often as it desired. What fantastic new dimensions this could offer to existence!

Expedition to Outer Space

"This initial contact with intelligent beings in outer space," continues the Director, "has made us revise our schedules for space exploration. We want to establish personal contact as soon as possible with this distant civilization. Even if we had a space ship that could travel at the speed of light, it would still take twenty-six years to get there. As you know, the fastest space ship we now have available was designed

to operate at a speed of 67,000,000 miles per hour, which is only one-tenth the speed of light. We've got a long way to go. I want plans drawn up within the next thirty days for gaseous nuclear reactor craft that can come within 90 per cent of the speed of light. We can use some of the spare capacity of Corcen in working this out.

"Einstein's theory of relativity indicates that when a spaceship travels around 90 per cent of the speed of light, time passes only about half as fast as it does on earth. Thus, our personnel will age only about fourteen and one-half years on a twenty-nine year journey at this speed. Everything aboard will seem normal, and this slowing of the clock will be noticed only after they return to earth. When we get our spaceships to travel within one-half of 1 percent of the speed of light, a year in space will produce the aging of about a month on earth. When our crew returns after a long voyage in space, any member will be considerably younger than, for example, an identical twin who remains on earth! We'll probably have little trouble getting volunteers," the Director says with a wry smile.

"We must make decisions soon regarding the life support systems. To be on the safe side, we should carry sufficient food and energy to last for a century. We must give thought to the type of personnel that will be best adapted for this trip. Should we attempt to arrest electronically most of the personnel aboard so that their bodies will not be subject to wear or deterioration during the time it will take them to get to this planet? To what extent should their human organs be replaced by the improved mechanical substitutes we now have available? Some of you may know that my heart and kidneys wore out about twenty years ago, and for the last two decades I've been living with a mechanical heart and kidney setup." At this point he taps his chest several times.

"I've never felt better, and they function perfectly. I believe if I were going on this voyage (at this point he looks a bit wistful), I would be better equipped because of my mechanical heart and kidneys. All medical teams here (Scott listens very carefully at this point) should be prepared within several months to give me their recommendations regarding the specifications for personnel to make this extended trip through space."

The Long-Range Program

The Director, who obviously has been under considerable excitement for several days, takes time to gulp down a container of protein drink spiked with a high concentration of water-soluble vitamins. After a few seconds he continues, "We should not let our space program be dominated by this single event. This is only the beginning, and we should think in long-range, overall terms. Let me review where we are today."

"As you all know," the Director says, "we first reached the moon around 1969, and within a decade several permanent stations were established there. The first permanent station was established on Mars in 1987. There are now over 10,000 people there. Venus took longer because of the 800 degree Fahrenheit surface temperature. We were able to utilize an enormous mountain that offered more comfortable temperatures. We have had an underground station on Venus since 2018. We will soon begin to cool the planet and oxygenate the atmosphere.

"Mercury presented us with interesting problems. It's nearest the sun, and it is approximately 3,200 miles

in diameter. We have a choice of a surface temperature of about 800 degrees on the part facing the sun or 400 degrees below zero on the part away from the sun. We've had an underground station there since 2026.

"It took us about a quarter-century after Mercury to work out the problems of Jupiter. We first landed on the largest moon, Ganymede. Temperature and radiation on the surface of Jupiter have not been as much of a problem as we expected, thanks to improvements in force-field technology. Jupiter has eleven times the diameter of earth and over a hundred times its area. The atmosphere is largely hydrogen and helium, and it is by far the stormiest of all the planets. Pressure was our biggest problem for no conventional space ship could withstand its crushing pressure that is approximated on our earth only at the bottom of our deepest oceans. Although this was one of the most hostile planets, we've had a colony of hardy scientists holed up on the South Pole of Jupiter for several decades, making valuable studies that have helped us in exploring the interior of the earth.

"Within ten days we expect to land on Titan, one of the moons of Saturn. Two years ago a manned space ship surveyed Uranus and its five moons. Our unmanned probes over the past century have given us vital information on both Neptune and Pluto. So much for the planets.

A Self-Sustaining Explorer

"We must no longer think in terms of our solar system. We have a universe to explore. We should begin work on an inter-galactic spacecraft engineered to leave this earth and never return!"

At this point there are muffled gasps throughout the room as the immensity of the conception breaks upon the audience. The room becomes quiet again. The Director continues.

"This spacecraft will be a sphere about a half-mile in diameter. It will carry 1,000,000 years' supply of nuclear energy, which will be replenished by absorbing radiant energy in space. The astronauts may get raw materials by "mining" space for asteroids and comets. The replicator aboard will enable them to convert energy to matter and also to convert matter to energy, whichever is needed.

"The spacecraft will probably have about 1,500 people aboard when it leaves us. Perhaps they will replicate people when more are needed. They will have craft for exploring and landing on unknown planets in the great reaches of space. What sort of personnel are best equipped to man this expedition toward infinity? Should we send humans? Perhaps we should redesign humans for survival in outer space*. Are humans hardy enough for the rigors of space? Should we send cyborgs—mechanical bodies with human brains? As you know, we have been successful in mechanically duplicating and improving every part of the human body, including the brain. Should we send mechanical men who have no fragile human parts? They would not be affected by radiation, below zero temperature, or lack of oxygen. They could accomplish hazardous jobs in space that would mean certain death to a human. They would have no food or elimination requirements. Energy for a century could be built in. These mechanisms would survive stress that would kill everyone else. They would be immortal; any part that might wear out could easily be replaced. Although mechanical men with these specifications are not available today, we expect to have them soon. Perhaps we should plan to use all three types on this expedition.

"This vanguard of our civilization will probably receive our signals for forty years. Because of the time lag we will probably hear from them for forty years after our signals have become too dim for them to pick up. Then these explorers will really be on their own—never again to communicate with us in any way. They may choose to branch out over the planets of the entire universe so that billions of years from now these children of earth will approach the outside of the universe—if it has an outside. There are probably over ten billion planets suitable for the birth and development of life as we know it. No matter what they do, people on earth will probably never know about it. Even if communication were possible, it would hardly be hot news by the time we received it." There are several chuckles throughout the room.

"Although we will never know where they are or what they are doing, we can be sure that they and their offspring colonies will be very busy. The diameter of the galaxy of flaming suns in which our earth is located is 100,000 light years across. As you know, light traveling through space at the rate of 186,300 miles per second will travel about six trillion miles in a year. Alpha Centauri, the nearest star beyond our sun, is approximately 25 trillion miles away. It takes light about four and a quarter years to reach us from this star. Our own galaxy contains one hundred billion stars with, we presume, countless planets on which life exists. And our galaxy is only one out of tens of billions of similar galaxies scattered throughout the expanding universe. It would be simpler to study every grain of sand on every beach in the world than to explore the universe.

"The uncharted seas of space are almost without limit. If we make an analogy with the explorations of Columbus, it would seem by comparison that this wily old explorer did little more than stick his big toe out the back door. The decisions that we make in the next few years will deeply affect the destiny of the human race and, who knows, perhaps the universe. Man has a way of cutting an ever-widening trail wherever he goes."

At this point the Director sits down, obviously filled with a deep sense of the interaction of present and future. The scientists gradually leave the room. There is no conversation. Everyone is engaged in his own thoughts. Scott wonders whether he can still meet Hella on the moon as they have planned.

15.The New Personality

Hella has tentative plans to meet Scott at the main celestial mechanics observatory on the moon. Scott, however, has become absorbed in space research that will play a part in the first face-to-face contact with extra-terrestrial beings. In the meantime, an opportunity has come to Hella that is too interesting to turn down. Corcen has checked with her regarding her interest in observing a group of people from the twentieth century that have been thawed out.

In the latter part of the twentieth century, to avoid the finality of death many people had themselves quick-frozen immediately before or after death. They hoped that by having their bodies preserved they could be thawed out at a later date with minimal damage so the medical skill of a future civilization could bring them back to life. One of the more dubious legacies of the past was about twenty-two thousand of these frozen people.

No one knows exactly what to do with these corpses. Should an attempt be made to resurrect them? Should they simply be disposed of? Since the population of the world is maintained at a constant level, most people feel that it is preferable to create a new life that is genetically and psychologically prepared for participation in the twenty-first century. Resurrection of one of these bodies with an uncertain adjustment in the twenty-first century might be a sticky business.

When this problem was referred to Corcen, a firm conclusion was returned in seconds: make no attempt at resurrection. The individuals of the twenty-first century deeply accept Corcen. They have found it reliable in 99.97 per cent of its predictions during the past eighteen years (it had inadequate data on the 0.03 per cent it missed). Yet, many individuals feel they can not ignore the human hopes that lay frozen in these modern catacombs.

Corcen does not dictate how things should be run in the tweny-first century, it only advises. As it perceives the thoughts of the people, it operates in a manner that fulfills their needs. It usually has better insight into what brings people happiness than any individual person. This has been proved time and time again by its successful predictions.

Nevertheless, people are free to do as they wish. Finally, a group decided they would attempt the revival of 100 of the bodies. They picked fifty males and fifty females whose records seemed to be especially promising and thawed them out. They have been successful in bringing 93 per cent of them back to life, and replacing the defective organs that were responsible for death with synthetic organs.

The real problem arose when they found that these individuals are completely out of touch with patterns of life in the twenty-first century. You could no more leave them on their own than you could turn a baboon loose in the middle of a research center. • They seem so full of hostilities and have ego motivations that are so alien to the twenty-first century that people have finally given up the task of trying to train them to fit into the new world. These "thawees" are so disruptive of the routines of life in the twenty-first century that the group that has brought them to life realizes they are saddled with a custodial problem. They are beginning to understand the types of pressures and twenty-four-hours-a-day watchfulness that burdened mothers in previous centuries.

Their reverence for human life does not permit them to refreeze these "unsane" individuals. They finally decide to set up a twentieth-century behavioral research laboratory on an isolated island and turn these people loose there. They provide the ninety-three men and women with every material resource requested and build a laboratory for psychologists and anthropologists to observe them. The thawees are free to set up their own social structure.

Hella flies down to the isolated Pacific island. The staff is most delighted to see her. Although they are equipped with all of the twenty-first-century aids to living, including three-dimensional color telecommunication with all parts of the world, they feel cooped up. Varying one's environment is a part of twenty-first-century living.

Their Pathetic Heritage

By means of monitoring pickups, the custodians are able to make a recording of most of the behavior of the twentieth-century thawees. One evening as Hella is watching them on the teleprojection screen, two men begin to quarrel. One man suspects that another man has attempted to obtain a sexual relationship with a woman he feels belongs to him. Although the woman protests that his suspicions are not correct, he slaps her in the face and hits her in the ribs so hard that it sends her sprawling across the room. The man with whom she has been accused of intimacy stands up and rushes toward the attacking man. A fight begins that lasts several minutes.

Neither Hella nor anyone in the group has ever seen anyone strike another person in anger. They watch, spellbound, as the fight continues. They have read that twentieth-century television showed fights and murders. They know little children in that society sometimes spent from four to eight hours a day watching such vicious programs and learning these folkways. Hella, however, has never seen any of these films. She knows they are available upon request from Corcen. She has just never been curious about such obscenity.

The man who started the fight seems to be losing. Blood is streaming from his nose. Suddenly, the jealous man picks up a metal bar and brings it down with a crunching impact on the head of the man who came to the woman's rescue. His legs crumple, and he slumps to the floor. He is dead in minutes. The custodians watch incredulously.

The murderer is locked in a room by two other thawees. The next day a court is set up with a lawyer who thinks he should go free and a lawyer who asks for his death. A judge is appointed, and a jury is selected. Although the custodians have read of these tribal customs, they have never had an opportunity to experience them emotionally. It seems almost impossible that human beings could behave in this manner.

After several hours of verbal courtroom ritual, the jury labels the man "guilty," and the judge informs him that the group will take his life. They tie the man's hands behind his back and put a rope around his neck. They pull him several feet off the ground and watch self-righteously while he chokes to death.

Most of the custodians who observe this ritual become physically sick and vomit. They keep the recorders working, but they turn off the screen and walk outside to take deep breaths of fresh air. As they look over the vast Pacific Ocean, they manage to overcome their feelings of nausea at this strange spectacle of man's inhumanity to man.

Acceptance of Death

This strong reaction has not been caused by a fear of death. The people in the twenty-first century regard death as a natural phenomenon and accept it when it comes. They put their energy into living fully while they are alive. Every resource of medical science is used to keep bodies functioning, but each individual calls a halt when he feels that physical deterioration has gone too far. When the torch of life has burned

brightly, they do not hesitate to pass it on to another. Each individual realizes that upon his death a new baby will be permitted to enter into the world. They don't fight this natural progression beyond a certain point. In the future immortality may be possible. But until then—no problem.

While they are breathing the fresh ocean air, Hella asks where the other frozen bodies are kept. She is informed they are in the Antarctic vault near the frozen animal specimens. Hella is sure that they will remain there for some time. Perhaps thousands of years in the future when aggressive behavior is only a vague, theoretical concept, an intrepid group might wish to thaw additional specimens to observe this phenomenon. It seems unlikely that these frozen bodies could ever be functioning citizens in a contemporary society. Each year the antiquated associations locked in their frozen brains become more and more inappropriate to the rapidly changing world.

Hella wants to share these vivid experiences with Scott as soon as she can. Although she enjoys the intimate company of many men, for years her closest feelings have been toward Scott because of the parallel depth of intellect and feeling they share. Soon after she has recovered from the shock of witnessing the dual taking of life, Hella contacts Scott on the space research satellite.

"This place is busier than a frying molecule," Scott tells her. "We've been asked to find other quarters for our research. Since much of my work requires a cold minimum gravity area almost free of atmosphere, I'm moving to the moon. Labs are being installed right now. Come on up and join me."

"It sounds wonderful," she says. "I guess my feelings are a little too tender to continue with these twentieth-century animals at first hand. I'd much rather read about them. If you could have seen their faces, Scott—the deep insecurity, the hatred, the fear in their eyes. I've got to talk to you and be close to you."

Hella immediately informs the custodians of her plans. She is completely open and does not attempt to deceive them by saying that she has to go to the moon to help Scott set up a research station. Although this is true, it would be impossible for her to deal with a fellow human on any basis other than the full truth of her feelings. She describes her feelings and her apparent limited tolerance at this time for further observation of these relics from the twentieth century. Everyone understands what she is talking about. Were it not for their self-imposed responsibility, they would go with her.

To the Moon

Hella explains to Corcen her emotional need to get to the moon rapidly to be with Scott. Corcen always recognizes the feelings of humans and organizes the resources of the new world to meet their needs. A craft is diverted to pick her up within minutes. She arrives at a South American spaceport within forty minutes of her talk with Scott. Within six hours she is on the moon.

Although most of the structures on the moon have been built underground to simplify life-support systems and give protection from meteorites, there are several observation rooms above the surface. These rooms are formed by six-inch-thick, transparent metal domes.

When he first sees Hella, Scott can tell that she has been through an unusual emotional experience, and he understands her need to talk. He obtains an observation room that is not going to be used during the evening. How good it feels to be together again! Although they have not missed each other—for their lives have been busy with fulfilling activities which they shared by teleprojection—they both feel an unusually keen delight in their reunion.

They snuggle into the same living contour chair. The sides of their bodies touch warmly, unhampered by clothing. As they look through the glass of the observatory into the night beyond, they can see the bright ball of the earth a quarter-million miles away. Europe, Africa, and part of Asia are visible.

They feel toward earth as they suspect people in earlier times may have felt toward their mothers. Here is the organism that had brought them into being through eons of evolutionary time.

Here is the organism that nurtured them and made them what they are. Although they can not see the sun, they compare their feelings about it to those that children in earlier times may have had about a father. The energy that moves everything in their lives may have come from the sun. Even the atoms that formed the earth some four billion years ago may have been an offshoot of the sun—something like the spermatozoa that fathers of previous times contributed to the absorbing sexual potentials of the mothers.

"I think I've learned a lot about myself and our society in the last few weeks," Hella confesses. "I had taken everything for granted. A person with normal vision never appreciates what his eyes mean to him. It's only when you come face to face with blindness that you understand the part that your eyes play in your life." Hella smiles a slight, tender smile."It must have been disturbing to see the folkways of the twentieth century," Scott says, trying to empathize with her.

"Oh, it was," answers Hella, appreciative that Scott is working toward her in his feelings. She knows she can always count on him. "I learned a lot, but I would never go through it again."

"I heard two people were senselessly killed, one by an individual and one by the group," says Scott. "Did you actually see it?"

"Yes."

"It's hard to believe such things could happen—and at the same time that I was busy on the satellite working to contact intelligent beings in outer space. Just think, the *same* world, the *same* time," Scott is beginning to feel deeply about the deaths.

Hella does not want to get into a spiral of feelings about the double murders. She feels a need to steer the conversation toward a deeper appreciation of what they have—to understand the present in the light of the past.

"When I was on the island with the thawees," Hella says, "several of them kept insisting on seeing a lawyer. They didn't believe me when I told them we have no use for law or lawyers. They wanted to know what we do with criminals. I explained that we have no criminals—that people in our society of abundance don't act aggressively toward others. You have to be insecure and afraid in order to harm others. They

told me it wouldn't work—that I didn't know anything about human nature.

"I tried to explain that our supplemental brains are imprinted to make us want to seek assistance if we feel uncomfortable or hostile. Apparently, in their society they had to capture hostile people like wild animals. And just listen to this, Scott. They caged them in jails! People would voluntarily get medical help if they had a physical ailment, but sometimes they wouldn't get psychiatric help before they had done things that hurt someone."

"I suppose the thing that impressed me most deeply about them," Hella continues, "was the way they were driven so fiercely by their ego needs. I guess the scarcity conditions that set man against man accentuated the king-size egos we developed in our long evolution from the jungle. In trying to meet their ego needs and develop a feeling of individual worth, they got too concerned with their status in the eyes of others. They tried to nourish their starved egos with silly things like mink coats and diamond cuff links. They seemed to care less about being successful in their own terms; they were far more concerned with the appearance of success in the eyes of other people.

"It seems to me that one of our greatest differences lies in how we view ourselves," comments Scott, reflecting on the problem. "Our ancestors, at least in the twentieth century areas where these people came from, didn't have strong inner standards that expressed their own individuality. They were far more concerned with their reputations than their characters. Those poor people were like rudderless ships blown by winds of fashion and storms of capricious opinion."

"They just couldn't live by their own internal standards." Hella feels compassion for these people and the tragedy of their lost happiness. "I think this probably began with their early conditioning. Right from the word 'go' they were dominated by their parents. They *had* to do what their parents told them, or they'd be punished or made to feel bad. During their helpless, impressionable childhoods, they developed the habit of not judging and feeling things for themselves. "Moma knows best. Daddy won't like this.' The first five years were crucial. As they grew up, they were never free of these personality patterns."

Hella sits up quickly. "That explains why they never came into their own! Now I can see why their feelings always remained sharply tuned to picking up the first signs of any possible rejection that would indicate what to think and how to act."

The Supreme Ethical Standard

"Of course, our present way of life is not without roots in the past," Scott says. "In ancient Greece, Socrates advocated our supreme ethical standard: *'Gnothi seauton,'* know thyself." Scott pauses long enough to sit up. "And Shakespeare said, 'This above all, to thine own self be true, and it must follow, as the night the day, thou canst not then be false to any man.' "

"In spite of the teachings of many of the great thinkers, most of our ancestors really didn't understand what it means to know yourself or be true to yourself," says Hella. "These were hollow words, not a way of life. They were certainly not ethical guides by which people could run their lives."

"When you evolve your own standards of growth, you can feel greater worth and dignity every day," Scott

continues intently. "Sometimes simply listening to a moving piece of music will enlarge your esthetic experience and make you feel you've grown. Reading will add to your store of knowledge to give this feeling of increased being. Simply sensing your own feelings and developing a deeper insight into the powers of your brain can add to the feeling of worth. When you make personal growth a way of life, a feeling of individual fulfillment comes automatically."

"Since our feeling of worth is within our own control," says Hella, "we have a security our ancestors lacked. We can give to other people in ways that they could seldom give. Our ancestors could usually be generous toward their immediate families.

But they didn't have the resources to be generous toward a larger group. They had to compete too strongly with other people. Individuals in a larger group would hurt them and would take advantage of them. They had to wrestle money from them, they had to fight them for position, power, and prestige."

"You're limited in adding to the happiness of others," Scott concludes, "unless your own life is fulfilling."

Love Without Jealousy

"In previous times there was never enough of anything— money, security, or love," Hella says. "People developed feelings of possessiveness. The murder I recently saw revolved around the desire of a man to possess a woman. He seemed to feel he owned her—that he could tell her how to live her life."

"How barbarous," says Scott. "I can't imagine anyone trying to hold love by force or threats. You hold love with an open palm, not a closed fist."

"Yes, but they couldn't feel that way," Hella replies. "Their jealousies were brought on by feelings of inferiority and insecurity. A man was afraid that if the woman he loved was with another man, she would find this other man more attractive and not come back to him."

"When you're with someone else, I'm glad." Scott's tone is warm. "I know you've found a relationship that adds to your life. When you're with me, I have the satisfying feeling that we're together simply because we want to be with each other. In the past if a man and woman loved each other and wanted an intimate type of companionship, society expected them to bind themselves with a legal arrangement called marriage. Of course, this was done to provide for children during their growth years. But can you imagine fettering love and companionship with legal rights and obligations?"

"Sounds awful. If you enjoy being with me, it's wonderful." Hella's hand touches Scott gently. "If our paths grow apart, we've found more satisfying relationships elsewhere. Either way, we're ahead."

"Our open ways of feeling about each other and our ethical standard of being true to ourselves are perhaps the greatest social inventions of mankind," Scott philosophizes. "They could be realized only to a limited extent in previous societies. Only today's cybernated world can permit their full flowering. Warped children grow into warped adults. People who labor under an inferiority complex can't fully enter into this new way of feeling and acting."

"In the past," says Hella, "children spent their first five years of life under conditions that gave them a permanent inferiority complex. No matter how worthy they later became, no matter how learned, no matter how much power or skill they acquired, they always felt inferior to some extent."

"Those who struggled hardest for power, such as the Napoleons or Hitlers, were usually short men who had been heavily structured by the authority of their parents in their early years," adds Scott. "In an attempt to fight off their inferiority complex, they developed what outwardly looked like a superiority complex. But inside there always remained a scared little boy, insecure, trembling, and afraid someone would find out what he really felt. As long as people had an inferiority complex, it was impossible for them to get a fully secure feeling of worth based on their own inner development."

The Obscene Past

"The only thing that made me laugh while I was observing the resurrected twentieth-century people was their warped standards of obscenity," says Hella with a remembering smile. "One of the men had a drawing of a man and woman engaged in intercourse. Every woman that saw it acted shocked, and this seemed to give the man a perverse delight. I understand back in Victorian times a nude figure of a woman was considered obscene. Later, mores in Western society were revised so that only representations of the act of love were classified as lewd."

"Idiotic," Scott explodes. "How could a picture of one of the most beautiful experiences in life ever be considered obscene! A drawing might be crude, yes. It might be untalented . . ."

Hella interrupts. "We have a different way of applying the label 'obscene.' I have just witnessed the most obscene thing —a man hitting a woman in the face, a man turning on another person to kill him, a social group choking him to death with a rope." She shudders.

"Anything that degrades, debases, or dehumanizes a human being is considered obscene today," generalizes Scott. "Our ancestors in the twentieth century had a tremendous amount of obscenity. They plastered it all over their magazines, television screens, newspapers, and books. Murders, race prejudices, wars, etc."

"Buchenwald, Auschwitz, Dachau—ovens that devoured live, screaming human beings. Grotesque piles of human corpses— these are obscene things that were shown to men, women, and children in 'civilized' countries in the latter part of the twentieth century."

"Perhaps one of the most obscene things of all was a man in an electric chair twitching and writhing as the shock of electricity burned through his body," Scott grimaces. "I think the most obscene word used in twentieth-century America was the word 'nigger.' But few people seemed to realize it. They felt that their most obscene words were four-letter symbols for sexual and eliminatory acts. They wallowed in the worst sort of corroding filth without knowing it. They made attempts to cleanse themselves of four-letter words that had little to do with the human spirit."

Separate Worlds of Men and Women

"Did the women really decorate themselves the way they appear in the old films?" Scott asks, as he comically pushes his hair to one side.

"You wouldn't believe how many hours they spent putting their hair in weird shapes. They painted their fingernails red. They painted their eyelids purple. They used chemicals to make their cheeks a light red and their lips a darker red. What's more, they seemed to feel better when their heels were three inches off the ground and their toes sharply pointed in a way that bore no resemblance to the shape of their feet. Both the men and women seemed obsessed with youth. They apparently felt they had been going downhill since the age of twenty. They did everything they could to put up a hopeless fight against aging."

"I'd hate to live with such sham, such artificiality," says Scott. "Men and women seemed to build separate worlds for themselves. I believe they even used separate bathrooms."

"They did," Hella agrees. "Little boys and little girls were trained in ways that were very different. A little girl was encouraged to be a 'young lady.' Her toys were often dolls and doll houses, furniture, and cooking implements. A boy was considered a sissy if he showed much interest in these things. He was given guns and cowboy outfits. A little girl was considered a tomboy if she ran too fast or yelled too loud. In a thousand subtle and not so subtle ways, a woman was molded into patterns known as 'feminine,' and a young boy was encouraged to be what they called 'masculine.' Since young boys and girls are neither masculine nor feminine, but are just human, this created stress on many individuals. Their cultural training kept men and women from sharing the deeper worlds of feeling.

"Their sexual do's and don'ts were incredibly complex. Scott, they had rules about everything. Often, no variety was culturally permitted; you had your choice of one if you were married or none if you weren't. Some cultures even had laws governing the sexual positions people could use. Many societies frowned on women who expressed their sexual desires; it wasn't 'ladylike.' And intimate relationships with those of the same sex were often taboo."

"People in the twentieth-century Western culture had deeply inculcated guilt feelings which kept them from achieving an intense and ecstatic perfection in sexual pleasures," Scott points out. "Often a sexual climax was largely a shallow physical experience."

"In a way, we care for sex both more and less than the thawees I observed." replies Hella. "It is a more profound experience for us. And yet if we do not have it, we are so engrossed in other dimensions of life that we don't miss it. It is, paradoxically, more keenly enjoyed and less keenly missed. I find that my sexual feelings usually become more satisfying as I get to know a person better. And yet I enjoy the variety of occasionally being with other men,"

Communication of Feelings

"One of the great differences between our way of life and theirs," Hella continues, "seems to He in the degree to which we communicate our feelings. We talk about everything. The thawees seemed ashamed of their feelings. They often repressed them and weren't even able to face their own feelings, much less the feelings of other people. They hid behind polite masks."

"Didn't Mark Twain say, 'Only the truth is good manners'?" interjects Scott.

"Even husbands and wives would go through their lives miles apart in their inner feelings," says Hella. "Since they were ashamed of so many of their feelings, they felt it would hurt their image to let someone else know how petty their feelings were. And yet the other person was tortured by motives that were equally petty. This stupid mutual shame seemed to keep them from talking to each other and reaching out to touch each other."

"I can't understand how this could have happened," says Scott. "I don't believe I've ever had a feeling that I was ashamed of. I've had feelings that I didn't consider desirable, but they went away as soon as I expressed them to someone. Because other people have always empathetically received all of the feelings I've ever expressed and were not threatened by them I don't believe I've ever accumulated any mental baggage. I live fully here and now. The dead past and the unborn future don't control me."

"I remember reading about a twentieth-century man so angry at his wife that for eighteen years he never spoke to her," Hella says. "They lived together and ate at the same table, but he never talked to her. Finally, they went to a psychiatrist, who urged the man to communicate. The first thing the husband said was, 'I don't want to talk about it'"

"This was extreme," Scott replies. "Few people, however, were able to express their feelings fully to any other human being. Sometimes a few managed to do this with counselors. But rarely were they able to do this with those that were nearest and dearest to them and with whom they most needed to communicate their feelings. Instead, they wore masks and assumed personalities they didn't have. They used words to hide their real selves, both from themselves and others."

"In the old competitive world," Hella points out, "it was too risky to expose one's inner thoughts. They were afraid that other people would judge them, would be overly helpful, would give them advice that was not wanted, would start diagnosing them and tearing them apart and telling them what to do, or would save up this information and use it against them later. It was rare to find anyone who could listen with his heart."

"Think of all the unhappiness that could have been prevented if they had realized that feelings can be managed just as automobiles and space ships can be controlled," says Scott. "They didn't realize that unwanted feelings are sent on their way when you talk about them. Isn't it marvelous how unpleasant feelings are eliminated when you fully express them, and pleasant feelings are increased when they're expressed!"

"Look," exclaims Hella, "we can just begin to see the Americas!"

As Scott looks toward the bright earth suspended above their horizon, he is aware of a gentle reflected light that outlines their figures. The broad Atlantic Ocean is stippled by light patches, these must be large areas covered by clouds. Could that tiny white dot above the earth be the space research satellite? It's hard to tell. But the backlighting on Hella's breast that stops just above the nipple is beautiful.

The New Character

"I believe I've developed a much deeper appreciation of our culture," says Hella. "Our satisfaction and happiness lie within our own control. We may never approach our ideals of self-knowledge and self-development, but we can make continual and satisfying, day-by-day, minute-by-minute progress. This is what we need to have a meaningful life. We live broad, wide lives with an enormous range of interests. Our world is so large."

"We're closer, both to ourselves and to others," says Scott, "Somehow without giving up our own individuality, we seem to develop at the same time a deeper and more profound relationship with others. The more we find ourselves, the more we transcend the boundaries of our own egos. We give more of ourselves in our emotional relationships with other people, and, yet, we also retain a deeper ability to live by our own standards and to remain the masters of ourselves."

"In previous centuries togetherness meant a giving up of individuality rather than a strengthening of it."

"Yes, I know what you mean, Hella. We enhance our togetherness and reinforce our individuality at the same time. It sounds contradictory, but it isn't."

"I think it's our ability to communicate with each other that enables us to be intellectually and emotionally naked—to have no pretenses," Hella reflects."I guess that's one reason we enjoy being physically naked, too. We feel completely loved and completely secure. We have no need to hide, either from ourselves or others."

Hella pauses and drops her head back to rest on Scott's arm.

"And our love is not motivated by need. We do not love just to make up for a deficiency in ourselves. When we offer love, it is as a gift—a kind of spontaneous reaching out."

Scott feels that Hella's mood is changing. There are longer pauses. She is watching the earth, the satellites, and the stars. She has obviously shared with him the vibrant thoughts that have filled her as a result of her experiences in the past weeks.

He feels her hand on his chest. He turns his head toward her. She is looking straight into his eyes. He has accepting feelings of love. The universe is cold and objective, but the bits of space and time that contain human beings are filled with warmth, security, and affection.

"So there's life in other parts of the universe. Well, good!" says Scott. "How could it be better than life right here!"

PART III. Looking Forward

16. Education for Change

We don't view our projection of the twenty-first century as a final blueprint, and you shouldn't either. It will serve its purpose if it gets intelligent people thinking about these problems. We hope you can improve our projection of future goals and the ways by which they may be achieved.

"We are now at the point," said anthropologist Margaret Mead, "where we must educate people in what nobody knew yesterday and prepare in our schools for what no one knows yet, but what some people *must* know tomorrow." Perhaps never in the history of mankind has it been so important that we know where we're going and how to get there. Humanity is no longer running- a two-bit show. There are over 3,000,000,000 people in the world today. We will soon be able to start a nuclear war that could wipe out all human life. Even ignoring the nuclear threat, it will take global organization of a high order to provide a good life for all humans. Freedom from war and want is at last within our grasp. But it won't happen automatically. We must use our heads and our hearts.

Scientific, political, industrial, economic, and sociological changes are occurring at a pace more rapid today than ever before in history. Some people have wished that things would slow down so we could have more time to adjust to change. This, of course, won't happen. Many are opposing changes simply because they are changes. They are nostalgically and frantically holding on to the "wisdom" of the past. But in times of rapid change, the "wisdom" of the past is usually of little help in meeting the problems of the present.

W. H. Ferry of the Center for the Study of Democratic Institutions has advised us that we should hardly be surprised by the changes that lie ahead:

Aristotle foresaw a takeover by machines 2000 years ago. The possibility of a workless or nearly workless society emerging from technology is part of our literature. H. G. Wells told his readers about it 50 years ago. Forty years ago, C. H. Douglas wrote: 'We can produce at this moment goods and services at a rate very considerably greater than the possible rate of consumption, and this production and delivery of goods can, under favorable circumstances, be achieved by the employment of not more than 25 per cent of the available labor working, let us say, seven hours a day.' Olaf Stapledon and Stuart Chase, in very different ways, told us the same story 30 years ago. Jacques Ellul in The Technological Society, just published, says, 'By the end of the 19th Century people saw in their grasp the moment in which everything would be at the disposal of everyone, in which man, replaced by the machines would have only pleasures and play.' In a neglected report of December 1963, the Research Institute of America anticipated the Committee when it remarked:

> *'The moment of truth on automation is coming—a lot sooner than most people realize. .The shattering fact is that the U.S. is still almost totally unprepared for the approaching crisis.'**

** W. H. Ferry, "Further Reflections on the Triple Revolution," Fellowship, January, 1965.*

It seems incredible that any intelligent man can view with complacency the slowness with which we are changing to meet the challenge of the world that lies ahead of us. Dandridge M. Cole has pointed out that,

> *"It has already been noted that technical knowledge is doubling every seven years (the doubling time is decreasing), and that ninety percent of all the scientists who ever lived are alive right now. Without assuming any reduction in doubling time it can be estimated that our total of technical knowledge in fifty years will exceed the present level by a factor of 2^7 or 256."**

**Dandridge M. Cole, Beyond Tomorrow (Amherst, Wisconsin: Amherst Press, 1965), pp. 87-90.*

"In the past most individuals were able to go through life with the set of attitudes and beliefs appropriate to the age in which they were brought up," wrote Robert Theobald:

> *The rate of change in science, in technology, in the beliefs and ideals of man was sufficiently slow to ensure that they remained relatively appropriate. Even then the older generation expressed its dissatisfaction in the phrase: 'I don't know what the world is coming to.' Nowadays it is recognized that the attitudes appropriate to the beginning of the twenty-first century will be totally different from those now accepted, but little attempt is made to look ahead. Indeed, much education is based on the ideas of past scholars; as a result theories are taught to generations of students long after they have been recognized by the leaders in the field of study to be incorrect.***

***Robert Theobald, The Rich and the Poor (New York: Clarkson N. Potter, Inc., 1960), pp. 139-140.*

Every kindergarten, elementary school, high school, and college in the nation should help students anticipate the changes that lie in their future. It should challenge them to seek new ways of thinking and feeling—of reorganizing their society to make the most of man's potential for happiness in the new age. Instead, most of our public and private schools prepare the students to live with the values and folkways of our ancestors.

Ready or Not

Ready or not, we are rapidly launching into a period of tremendous change. This is obvious on a technological level with satellites orbiting the earth, color television in our homes, and the government computers checking up on our income tax. But we are just now beginning an era in which social change must keep pace with technological change. The social patterns which we have inherited from ancient Mesopotamia will not give us happiness in the world of the future. The turmoil, insecurity, unhappiness, and conflict that are experienced today will increase unbearably if we are slow in inventing new ways of living, thinking, and feeling. Humanity is now entering into its adolescent phase. If we're going to come through our teen years without too many scars, the human race had better learn how to mature.

Perhaps the greatest threat that faces us at this moment is the fragmentation of humanity into over 100 egocentric national boundary lines. These paranoid nationalities claim the sovereign right to use weapons that can kill millions of people in other countries. If we continue improving our atomic weapons for another twenty years, it is possible that a flare-up of a dictator's temper could bring on a chain of events that would wipe out every human being.

No one can predict the future with certainty. One thing, however, seems highly probable. Things are moving so fast that in a hundred years our society will bear little resemblance to the economic, social, and political patterns of today. We suggest that whatever the future brings, it will represent a pattern spun out by the "Life, Liberty, and the Pursuit of Happiness" value structure, the scientific method as a thinking technique, and the cornucopia flowing from automated and computerized tools.

The future holds great stress and threat to individuals who do not have flexible nervous systems. It also offers a limitless challenge to those who can use their intelligence for its primary evolutionary function: to adapt to changing conditions. Greater wisdom, fantastic accomplishments, and enormously increased happiness may be ours in the humanistic, scientific, cybernated world of tomorrow.

Your Participation Is Needed

"In world leaders and individual citizens alike," advises Dr. Robert M. Hutchins,

*"old habits and customs stand in the way of adaptation to a new world. We are only beginning to study these habits and customs, to seek new ways of using our intelligence in order to preserve the species. In this effort the best thinking of every man and woman is needed."**

> **Change, (Santa Barbara, California: Center for the Study of Democratic Institutions, February, 1965), Vol. 1, No. 1, p. 1.*

Dr. George Gallup in *The Miracle Ahead* points out that we can not rely on our economic and political leaders to help us respond dynamically to the challenge of the future. Dr. Gallup suggests:

> *. . . change can not be brought about easily by its leaders, except in those situations in which the changes advocated do not disturb present relationships. In fact, it is the leaders who typically become the most bitter and the most effective foes of change. The public, therefore, must take the initiative and assume responsibility for progress in the affairs of man. The public must force change upon its leaders.***

> *** George Gallup, The Miracle Ahead (New York, Evanston, and London: Harper & Row, 1964), p. 201.*

In the history of man, no generation has been educated to expect social change and creatively adapt to it. In a very real sense we back into the future hoping we don't get our behinds chewed off. We eagerly seek new medicines for our physical ailments even before these have been thoroughly tested. But when it comes to political, social, and economic changes upon which so much of our happiness depends, our guiding philosophy seems to be, "Don't rock the boat." Well, the boat is rocking, and it's going to remain rocking for a long time. The only way to stop the boat from rocking is to use the scientific method of thinking to guide us to social inventions that really work.

No one today has all the answers—or even the questions. But by means of careful experimentation and measurement of results, we can eventually determine what political, economic, and social changes will give humanity freedom from war and want and will enable all people to live a more satisfying life.

For the first time in the history of man we can redesign both ourselves and our entire environment! By manipulating our genes, we will be able to change the structure and function of our bodily organs in almost any way that we desire. With a cybernated technology based on nuclear power, we can redesign our living areas, our cities, and our planet. And even the sky is not the limit. Eventually, the mind of a man might profoundly change the planets of our solar system. Our galaxy and possibly the nebulae beyond may feel our touch. Our only limitations are our intelligence and our creative imagination. *Man may now control his destiny!*

Every intelligent citizen on this globe should be pondering these uncharted seas on which the ship of humanity is now plunging ahead at full speed. Like Columbus, who set out on a daring voyage about a half-millennium ago, we have only a few scraps of information plus our scientific and humanistic intelligence to guide us. We must use these to the best of our ability to avoid the hell of atomic war. We must somehow land ourselves on a new shore where men and women may find themselves, where blighted personality development will become the exception rather than the norm, where man's inhumanity to man will be unknown, and where wars and want will be but a distant memory. Only then will the spirit of man soar to its full, wonderful potential.